

RENNES BRETAGNE AP, France

WMO: 071300

Lat: 48.0689N

Lon: 1.7339W

Elev: 38

StdP: 100.87

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.5	-2.1	-6.9	2.1	-1.3	-5.0	2.5	-0.5	12.3	10.5	11.1	10.2	1.9	20	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	30.3	20.1	28.1	19.2	26.1	18.5	21.2	28.0	20.1	26.1	19.2	24.4	3.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	23.7	18.0	13.0	22.2	17.2	12.3	21.3	61.5	28.0	57.6	26.1	54.6	24.6	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	8.4	7.4	-6.0	34.1	1.8	2.2	-7.3	35.7	-8.3	37.0	-9.3	38.2	-10.6	39.8
			-6.5	22.8	1.9	1.4	-7.8	23.8	-9.0	24.6	-10.0	25.4	-11.4	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	6.0	6.5	8.6	10.6	13.9	17.1	19.1	18.9	16.3	13.2	9.0	6.4
	DBStd	5.68	3.74	3.31	3.00	2.96	3.06	3.14	2.76	2.85	2.76	3.03	3.17	3.86
	HDD10.0	509	129	103	63	27	3	0	0	0	0	8	55	121
	HDD18.3	2380	382	332	302	231	141	59	23	26	73	161	281	371
	CDD10.0	1302	6	5	19	46	125	214	282	275	190	106	24	9
	CDD18.3	130	0	0	0	0	4	23	47	43	12	1	0	0
	CDH23.3	1233	0	0	0	8	51	246	440	368	115	6	0	0
	CDH26.7	368	0	0	0	0	6	77	137	123	24	1	0	0
Wind	WSAvg	3.6	4.0	4.0	3.9	3.7	3.6	3.4	3.4	3.2	3.2	3.3	3.5	3.7
Precipitation	PrecAvg	753	76	58	51	57	61	51	52	48	59	78	83	79
	PrecMax	966	178	142	164	162	129	139	135	116	166	139	167	184
	PrecMin	577	9	3	7	9	22	8	12	8	8	23	30	18
	PrecStd	122	43	31	34	34	26	32	28	30	40	33	36	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	15.9	20.5	24.6	27.4	32.9	33.0	33.8	29.8	23.7	17.3	14.7
		MCWB	12.4	11.4	13.3	15.7	18.3	21.0	20.7	20.9	19.1	17.3	15.0	13.2
	2%	DB	13.2	13.5	17.3	21.1	24.6	28.7	30.1	29.9	26.6	21.0	15.9	13.9
		MCWB	11.7	10.9	12.0	14.7	17.0	19.4	19.7	20.1	18.1	16.7	14.0	12.6
	5%	DB	12.2	12.3	15.2	18.6	22.4	26.0	27.8	27.2	24.1	19.1	14.8	13.0
		MCWB	10.9	10.3	11.3	13.0	16.1	18.4	19.1	19.3	17.2	15.6	13.2	11.9
	10%	DB	11.3	11.5	13.6	16.4	20.0	23.7	25.6	24.8	22.0	17.8	13.8	12.0
		MCWB	10.0	9.8	10.7	11.7	14.7	17.3	18.4	18.1	16.3	15.2	12.2	10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.5	14.1	17.0	19.7	22.8	22.2	22.7	20.5	18.5	15.9	13.8
		MCDB	13.9	14.3	18.2	22.4	25.6	30.4	30.3	31.0	26.4	21.8	16.7	14.2
	2%	WB	12.1	11.6	12.8	14.9	17.9	20.4	20.9	21.2	19.1	17.3	14.5	12.9
		MCDB	13.1	13.0	15.7	20.1	23.0	26.8	27.9	27.8	24.4	19.9	15.5	13.6
	5%	WB	11.1	10.8	12.0	13.5	16.5	19.0	19.8	20.0	18.1	16.5	13.4	12.1
		MCDB	12.1	12.1	14.5	17.5	21.3	24.8	25.8	25.3	22.4	18.5	14.6	12.9
	10%	WB	10.2	10.0	11.2	12.3	15.3	17.7	18.9	18.9	17.1	15.6	12.4	11.1
		MCDB	11.2	11.4	13.3	15.7	19.3	22.6	24.1	23.3	20.7	17.4	13.6	12.0
Mean Daily Temperature Range	5% DB	MDBR	5.8	7.1	8.5	9.7	9.9	10.5	10.8	10.8	10.4	8.1	6.7	5.9
		MCDBR	5.6	7.7	11.2	13.6	13.6	14.5	15.1	14.7	13.8	9.3	6.9	5.5
		MCWBR	4.9	5.7	6.8	7.5	7.0	6.7	6.2	6.0	6.4	5.5	5.3	5.0
	5% WB	MCDBR	5.5	6.8	9.2	11.8	12.1	13.1	13.0	12.4	11.3	8.0	6.3	5.4
		MCWBR	5.1	5.6	6.0	6.9	6.6	6.6	5.9	5.7	6.0	5.4	5.2	5.1
Clear-Sky Solar Irradiance	taub	0.330	0.346	0.376	0.393	0.400	0.399	0.396	0.385	0.367	0.361	0.336	0.323	
	taud	2.410	2.373	2.285	2.244	2.271	2.308	2.321	2.371	2.400	2.417	2.422	2.386	
	Ebn at Noon	734	805	834	856	863	864	861	856	836	777	719	689	
	Edh at Noon	68	87	111	127	129	125	122	111	98	83	67	62	
All-Sky Solar Radiation	RadAvg	1.02	1.85	2.96	4.43	5.24	5.80	5.54	4.82	3.81	2.20	1.26	0.87	
	RadStd	0.10	0.21	0.33	0.49	0.43	0.59	0.35	0.40	0.29	0.17	0.12	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page